3.3 Backward Error Analysis for Hamiltonian Splitting Methods

As the relation

$$\mathcal{L}_H \phi = \{\phi, H\}$$

is linear in H, we may write, for Hamiltonians H_1, H_2,

$$\mathcal{L}_{H_1+H_2} = \mathcal{L}_{H_1} + \mathcal{L}_{H_2}.$$

The flow map of the system with Hamiltonian $H = H_1 + H_2$ is

$$\mathscr{F}_t = e^{t(\mathcal{L}_{H_1}+\mathcal{L}_{H_2})}.$$

On the other hand, the splitting method based on a composition of flows on H_1 and H_2 is

$$\mathscr{G}_h = e^{h\mathcal{L}_{H_1}} e^{h\mathcal{L}_{H_2}}.$$

The issue then becomes the comparison of the exponential series

$$e^{h(A+B)}$$

with

$$e^{hA} e^{hB},$$

for noncommuting operators A, B. Expanding these out using the exponential series we have

$$\begin{aligned} e^{h(A+B)} &= \mathrm{Id} + h(A+B) + \frac{h^2}{2}(A+B)^2 + \frac{h^3}{6}(A+B)^3 + \mathcal{O}(h^4) \\ &= \mathrm{Id} + h(A+B) + \frac{h^2}{2}(AB + BA + A^2 + B^2) \\ &\quad + \frac{h^3}{6}(A^3 + A^2B + AB^2 + ABA + B^2A + BA^2 + BAB + B^3) \\ &\quad + \mathcal{O}(h^4), \end{aligned}$$